JavaScript is disabled on your browser.

compact1, compact2, compact3

java.io

Interface Externalizable

All Superinterfaces:

Serializable

All Known Subinterfaces:

RemoteRef, ServerRef

All Known Implementing Classes:

ActivationDataFlavor, DataFlavor, MimeType, MLet, PrivateMLet

---

```
public interface Externalizable
extends Serializable
```

Only the identity of the class of an Externalizable instance is written in the serialization stream and it is the responsibility of the class to save and restore the contents of its instances. The writeExternal and readExternal methods of the Externalizable interface are implemented by a class to give the class complete control over the format and contents of the stream for an object and its supertypes. These methods must explicitly coordinate with the supertype to save its state. These methods supersede customized implementations of writeObject and readObject methods.

Object Serialization uses the Serializable and Externalizable interfaces. Object persistence mechanisms can use them as well. Each object to be stored is tested for the Externalizable interface. If the object supports Externalizable, the writeExternal method is called. If the object does not support Externalizable and does implement Serializable, the object is saved using ObjectOutputStream. When an Externalizable object is reconstructed, an instance is created using the public no-arg constructor, then the readExternal method called. Serializable objects are restored by reading them from an ObjectInputStream.

An Externalizable instance can designate a substitution object via the writeReplace and readResolve methods documented in the Serializable interface.

Since:

JDK1.1

See Also:

ObjectOutputStream, ObjectInputStream, ObjectOutput, ObjectInput, Serializable

Method Summary

Abstract Methods

Modifier and Type	Method	Description
void	**readExternal**(**ObjectInput** in)	The object implements the readExternal method to restore its contents by calling the methods of DataInput for primitive types and readObject for objects, strings and arrays.
void	**writeExternal**(**ObjectOutput** out)	The object implements the writeExternal method to save its contents by calling the methods of DataOutput for its primitive values or calling the writeObject method of ObjectOutput for objects, strings, and arrays.

Method Detail

writeExternal

```
void writeExternal(ObjectOutput out)
            throws IOException
```

The object implements the writeExternal method to save its contents by calling the methods of DataOutput for its primitive values or calling the writeObject method of ObjectOutput for objects, strings, and arrays.

Parameters:

`out - the stream to write the object to`

Throws:

`IOException - Includes any I/O exceptions that may occur`

readExternal

```
void readExternal(ObjectInput in)
           throws IOException,
                  ClassNotFoundException
```

The object implements the readExternal method to restore its contents by calling the methods of DataInput for primitive types and readObject for objects, strings and arrays. The readExternal method must read the values in the same sequence and with the same types as were written by writeExternal.

Parameters:

`in - the stream to read data from in order to restore the object`

Throws:

`IOException - if I/O errors occur`

`ClassNotFoundException` - If the class for an object being restored cannot be found.

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms. Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.